# Study 5G NR Architecture

**Radio Access Network:** The Radio Access Network provides a wireless connection between the user devices and 5G core Network.

1) **gNB**: The gNB is the 5G base station that provides radio connectivity between the User Equipment and the 5G core.

   **Functions:**

   - Provides wireless communication
   - Allocates radio resources
   - Performs beamforming
   - Handles handovers between neighboring cells
   - Manages scheduling and radio transmissions

   The gNB is equivalent to the 4G eNodeB but offers significantly improved performance and supports advanced 5G features.

2) **UE (User Equipment)**: User Equipment (UE) refers to any device that connects to the 5G network.

   **Examples:**

   - Smartphones
   - Tablets
   - Laptops
   - IoT devices
   - Smart Sensors

   **Functions:**

   - Connects to the 5G network
   - Sends and receives data
   - Performs authentication with the network

- Maintains mobility while moving between cells
- Requests internet and other network services

**<u>5G Core Network:</u>** The 5G Core (5GC) is responsible for authentication, session management, mobility, and data forwarding.

1) **AMF (Access and Mobility Management Function)**: The AMF manages signaling and mobility for users.
   **Responsibilities:**
   - User registration
   - Authentication
   - Security management
   - Mobility management
   - Connection management

   Whenever a mobile device powers on or moves between cells, the AMF manages the signaling procedures.

2) **SMF (Session Management Function)**: The SMF manages data sessions.
   **Responsibilities:**
   - Creates PDU sessions
   - Assigns IP addresses
   - Controls Quality of Service (QoS)
   - Selects the appropriate UPF
   - Manages user connectivity

   The SMF acts as the session controller of the network

3) **UPF (User Plane Function):** The UPF handles the forwarding of user data packets.
   **Responsibilities:**

- Packet routing
- Traffic forwarding
- QoS enforcement
- Traffic filtering
- Internet connectivity

The UPF is responsible for carrying the user's actual internet traffic.

**Interfaces in 5G:** Different network functions communicate using standardized interfaces.

| Interface | Connected Components | Purpose |
|---|---|---|
| **N1** | UE ↔ AMF | Registration, authentication, NAS signaling |
| **N2** | gNB ↔ AMF | Control plane signaling and mobility management |
| **N3** | gNB ↔ UPF | User data transfer |
| **N4** | SMF ↔ UPF | Controls user sessions and forwarding rules |
| **N6** | UPF ↔ Data Network | Internet and external network connectivity |

**5G Architecture Block Diagram:**

Data Network

(Internet/Cloud)

|

N6

|

UPF

User Plane Function

|

N4

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SMF

Session Management

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AMF

Access & Mobility

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N2

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GNB

5G Base Station

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N1

|

UE

Smartphone/IoT

**<u>5G Registration Call Flow:</u>**

```
UE                          gNB                         AMF
|                            |                            |
|-----Registration---->|                            |
|                            |----N2 Message----->|
|                            |                            |
|<------------Authentication-------------------|
|                          |                              |
|---------Authentication Response--------->|
|                         |                               |
|<------Registration Accept------------------|
|                        |                                |
|---------Registration Complete------------>|
```

**<u>Data Session Flow:</u>**

```
UE
|
               | User Data
gNB
|
```

|

AMF

|

|

SMF

|

|

UPF

|

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Internet/Data Network